

RAGAV U

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LinkedIn | GitHub | Portfolio

SUMMARY

Third-year B.Tech student in Artificial Intelligence and Data Science with hands-on experience in Python, Java, NLP, computer vision, and agentic AI systems. Built full-stack AI projects using FastAPI, React, and Supabase, including a real-time Indian Sign Language recognition system and a multilingual RAG assistant for Indian IP law. Shortlisted among the top 50 teams in the SIH 2026 Internal Hackathon. Seeking an entry-level Software or AI/ML Engineer role.

EDUCATION

B.Tech, Artificial Intelligence and Data Science 2024 – 2028 (Expected)
V.S.B. Engineering College, Karur, Tamil Nadu | CGPA: 8.22 / 10.0 (Semester 5)
Coursework: Data Structures and Algorithms, Database Management Systems, Machine Learning, Natural Language Processing, Computer Vision, Deep Learning, Operating Systems, Distributed Computing, Object-Oriented Programming

TECHNICAL SKILLS

Languages	Python, Java, SQL
AI / ML	Scikit-learn, XGBoost, Random Forest, Gradient Boosting, SHAP, TensorFlow (basics), OpenCV, MediaPipe, NLTK, TF-IDF, Pandas, NumPy
GenAI & Agents	Gemini API, RAG, Agentic AI, Multi-Agent Systems (Google ADK), pgvector
Backend & Web	FastAPI, Flask, Spring Boot, REST APIs, React, TypeScript, JWT Authentication, Supabase
Databases	MySQL, PostgreSQL (Supabase), pgvector
Speech & NLP	gTTS, pyttsx3, Speech Recognition, Tokenization, Text-to-Speech
Tools	Git, GitHub, Render, Visual Studio Code, Jupyter Notebook
Core CS	Data Structures and Algorithms, OOP, DBMS, Operating Systems

PROFESSIONAL EXPERIENCE

Virtual Intern 2026 – Present
Infosys Virtual Internship 7.0 (Remote)

- Currently enrolled in the Infosys Virtual Internship 7.0, working through structured, industry-oriented learning modules and project assignments.
- Applying Python, SQL, and AI/ML fundamentals to guided project tasks following industry software development practices.
- Following milestone-based deliverables and mentor guidance in a self-managed remote workflow.
- Building professional skills in documentation, teamwork, and structured problem solving.

Artificial Intelligence Intern Jan 2026 – Feb 2026
Enginow (Remote) — Noida, Uttar Pradesh

- Completed a structured 4-week AI development program covering supervised machine learning, NLP fundamentals, and Python automation in a remote team environment.
- Debugged and optimized Python data processing scripts, reducing runtime by about 25% through algorithmic improvements and code refactoring.
- Took part in weekly code reviews with senior developers, gaining exposure to Git workflows, industry coding standards, and agile practices.
- Delivered all assigned milestones on schedule in a self-managed remote workflow.

PROJECTS

Vision-Speak: Sign Language to Speech with Emotion Detection

Personal Project | 2025 – Present

Python, OpenCV, MediaPipe, TensorFlow, gTTS, NLP | [GitHub](#)

- Built a real-time system that detects Indian Sign Language gestures using MediaPipe hand landmarks and an MLP classifier trained on custom gesture data.
- Implemented a text-to-speech pipeline with gTTS that converts recognized gestures into spoken Tamil and Hindi, improving accessibility for the hearing-impaired.
- Integrated an emotion detection module (OpenCV, FER) that displays the detected emotional state alongside the sign translation in real time.
- Designed an end-to-end pipeline of frame capture, hand landmark extraction, gesture classification, NLP text processing, and regional-language speech output, running at 24+ FPS.
- Targets accessibility for roughly 1.8 million hearing-impaired people in India who use ISL as their primary language.

IP - Sakthi Sahayak: Multilingual RAG Assistant for Indian IP Law

Hackathon Project | SIH 2026

Python, FastAPI, Java, Spring Boot, React, TypeScript, PostgreSQL, Gemini API | [GitHub](#)

- Developed a RAG-based legal intelligence assistant for Indian Intellectual Property law, grounding every answer in a corpus of 25 verified documents (7,019 chunks).
- Implemented hybrid retrieval with a legal reranker, citation validation, and abstention on out-of-scope queries; reached 55/55 on the runtime evaluation set with an MRR of 0.97.
- Built a three-layer architecture: FastAPI RAG service, Spring Boot orchestration with JWT authentication, and a React frontend with voice chat and evidence cards.
- Added multilingual support (34+ languages) with speech input and output; led the team and prepared the jury presentation.

SolarShield AI: Solar Flare Detection

Personal Project

Python, XGBoost, SHAP

- Developed a solar flare detection model using XGBoost with SHAP explainability on ISRO Aditya-L1 mission data.

HACKATHONS AND ACHIEVEMENTS

- **Top 50 teams, SIH 2026 Internal Hackathon** (Team Leader): problem statement on an AI multilingual, RAG-based assistant for intellectual property and regulatory guidance.
- **Kaggle / Google ADK Hackathon**: built StudyMate AI, a multi-agent academic concierge, after completing the Kaggle 5-Day AI Agents Intensive.
- **HackerRank Orchestrate (June 2026)**: participant. Active participant in multiple hackathons and technical events across Tamil Nadu.
- **Competitive programming**: 100+ LeetCode problems solved in Java (arrays, HashMaps, two-pointer, sliding window, tree traversal).
- **Languages**: fluent in Tamil and English, enabling regional-language AI solutions such as Tamil speech output in Vision-Speak.

CERTIFICATIONS

- Gemini Enterprise Skill Badge — Google Cloud
- 5-Day AI Agents Intensive — Kaggle
- GenAI-Powered Data Analytics Job Simulation — Tata Group via Forage (2026)
- Data Analytics Job Simulation — Deloitte via Forage (2026)
- Data Fundamentals — IBM (2026)
- Introduction to AI Assistants: Claude 101 — Anthropic (2025)